ANTENNA DESIGN Considerations For RF Applications

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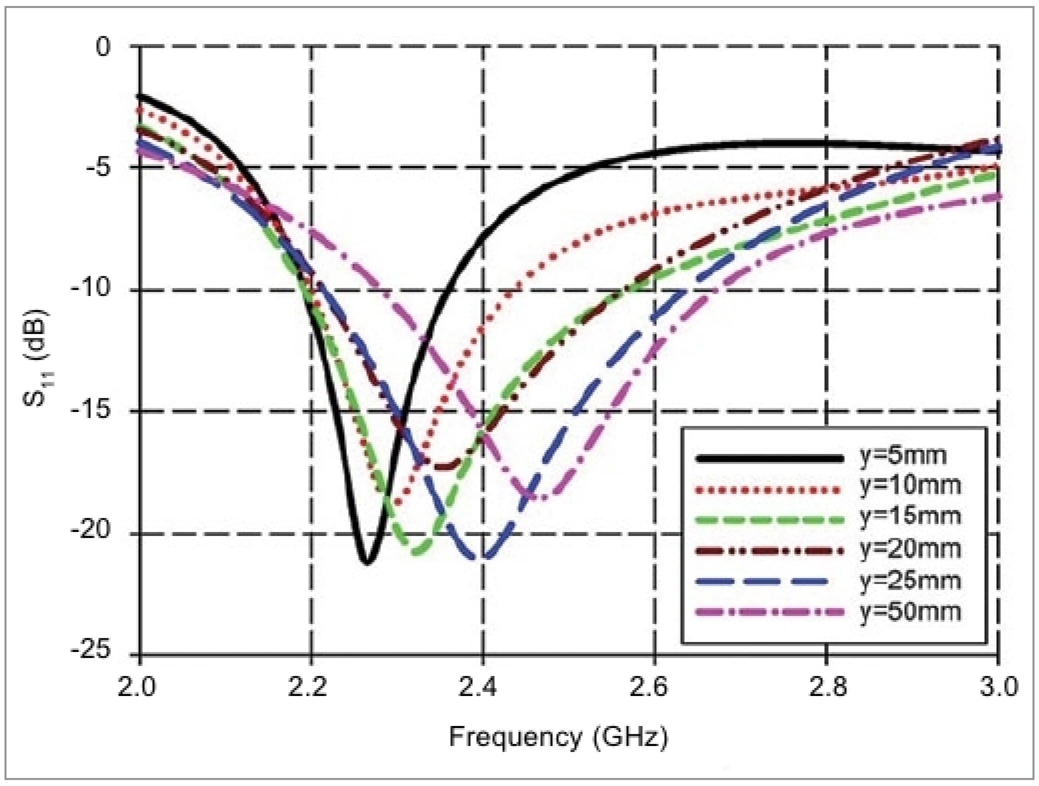
Since there is no unique solution for a particular RF application, some guidelines are needed. This article looks at some such guidelines

An antenna transmits and receives electromagnetic (EM) radiation in free space. The wireless range of an antenna depends greatly on its design, enclosure and a good PCB layout. This article covers some of the best practices for antenna design for radio frequency (RF) applications, to get the widest range possible with a given amount of power.

Other general design considerations include antenna length and feed, shape and size of ground plane, and return path. Selection of an antenna depends on application, available board size, RF range, directivity and cost. For example, a Bluetooth mouse works in 30cm range and requires a data rate of only a few kbps, while a voice-controlled device requires 60cm range and a data rate of 64kbps.

Chip antennae are a good solution for applications where PCB size is extremely small. However, these increase the bill of material (BOM) and assembly expenses. These are sensitive to RF ground size, and require an additional matching network for antenna tuning, as the same cannot be achieved by changing antenna length. This further increases BOM cost.



Effects of ground plane length on resonant frequency (Credit: Research Gate)

Effects of enclosure and ground plane

An antenna operating frequency is inversely proportional to inductance (L) or capacitance (C). These parameters are sensitive to PCB RF ground size and the product's plastic casing. A large RF ground plane and plastic casing increase effective capacitance. Plastic has a higher dielectric constant than air, so proximity of plastic to the antenna results in a higher effective dielectric constant. This also increases the electrical length of antenna trace, reducing the resonant frequency of the antenna.

For a good PCB layout, the better the ground provided for a quarter-wave antenna, the better it correlates with theoretical behaviour. Return path of the RF signal is in the ground plane beneath RF trace. For good RF performance, return path should be uninterrupted and as wide as possible. If ground plane is interrupted, return currents find the next smallest path around the interruption. This forms a current loop, adding undesired inductance, affecting impedance match between the radio and antenna, consequently, attenuating RF signal significantly. If ground plane beneath the RF trace is narrow, it does not behave like a microstrip and may have more signal leakage.

To maximise radiation in the desired direction, orientation of the antenna should be in line with the final product's orientation. It is important for an antenna to match the network, because a lot of parameters in the antenna's proximity can vary its impedance. Also, it